

Generalization of a Riemann Zeta Function Series

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Abstract

We present a rigorous evaluation of the generalized series $\sum_{x=m}^{\infty}(\zeta(x) - 1)$ for integers $m \geq 2$, where $\zeta(x)$ denotes the Riemann zeta function. We demonstrate that the infinite sum collapses into a finite expression involving integer combinations of zeta values.

1 Introduction and Theorem

The image under consideration evaluates the limit:

$$\lim_{n \rightarrow \infty} \sum_{x=2}^n (\zeta(x) - 1) = \sum_{x=2}^{\infty} (\zeta(x) - 1) \quad (1)$$

By unpacking the definition of the Riemann zeta function, we can generalize the lower bound of this summation to an arbitrary integer $m \geq 2$.

Theorem 1. *For any integer $m \geq 2$, the following identity holds:*

$$\sum_{x=m}^{\infty} (\zeta(x) - 1) = (m - 1) - \sum_{j=2}^{m-1} \zeta(j) \quad (2)$$

where it is understood that the empty sum (when $m = 2$) evaluates to 0.

2 Proof of the Theorem

Proof. Recall the series definition of the Riemann zeta function for real $x > 1$:

$$\zeta(x) = \sum_{k=1}^{\infty} \frac{1}{k^x} \implies \zeta(x) - 1 = \sum_{k=2}^{\infty} \frac{1}{k^x} \quad (3)$$

Substitute this into the original sum:

$$S_m = \sum_{x=m}^{\infty} (\zeta(x) - 1) = \sum_{x=m}^{\infty} \sum_{k=2}^{\infty} \frac{1}{k^x} \quad (4)$$

Since all terms $\frac{1}{k^x}$ are strictly positive for $k \geq 2$ and $x \geq 2$, the Tonelli theorem for discrete sums allows us to interchange the order of summation without altering the result:

$$S_m = \sum_{k=2}^{\infty} \sum_{x=m}^{\infty} \left(\frac{1}{k} \right)^x \quad (5)$$

The inner sum is a convergent geometric series with first term $\frac{1}{k^m}$ and common ratio $\frac{1}{k} < 1$:

$$\sum_{x=m}^{\infty} \left(\frac{1}{k}\right)^x = \frac{k^{-m}}{1 - k^{-1}} = \frac{1}{k^{m-1}(k-1)} \quad (6)$$

Thus, the sum becomes:

$$S_m = \sum_{k=2}^{\infty} \frac{1}{k^{m-1}(k-1)} \quad (7)$$

To evaluate this series, we employ a partial fraction decomposition on the summand. We seek coefficients $A, B_1, B_2, \dots, B_{m-1}$ such that:

$$\frac{1}{k^{m-1}(k-1)} = \frac{A}{k-1} + \sum_{j=1}^{m-1} \frac{B_j}{k^j} \quad (8)$$

Multiplying both sides by $k^{m-1}(k-1)$ yields:

$$1 = Ak^{m-1} + (k-1) \sum_{j=1}^{m-1} B_j k^{m-1-j} \quad (9)$$

Setting $k = 1$ immediately gives $A = 1$. Setting $k = 0$ gives $B_{m-1} = -1$. By comparing the coefficients of the highest power of k (which is k^{m-1}), we find $A + B_1 = 0$, thus $B_1 = -1$. Iteratively comparing descending powers of k reveals that $B_j = -1$ for all $1 \leq j \leq m-1$. Hence:

$$\frac{1}{k^{m-1}(k-1)} = \frac{1}{k-1} - \sum_{j=1}^{m-1} \frac{1}{k^j} = \left(\frac{1}{k-1} - \frac{1}{k} \right) - \sum_{j=2}^{m-1} \frac{1}{k^j} \quad (10)$$

Substitute this decomposition back into (7):

$$S_m = \sum_{k=2}^{\infty} \left[\left(\frac{1}{k-1} - \frac{1}{k} \right) - \sum_{j=2}^{m-1} \frac{1}{k^j} \right] \quad (11)$$

Since the component series are absolutely convergent, we can split the limits:

$$S_m = \sum_{k=2}^{\infty} \left(\frac{1}{k-1} - \frac{1}{k} \right) - \sum_{k=2}^{\infty} \sum_{j=2}^{m-1} \frac{1}{k^j} \quad (12)$$

The first sum is a standard telescoping series that evaluates strictly to 1:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots = 1 \quad (13)$$

For the second sum, interchanging the finite sum over j with the infinite sum over k yields:

$$\sum_{j=2}^{m-1} \sum_{k=2}^{\infty} \frac{1}{k^j} = \sum_{j=2}^{m-1} (\zeta(j) - 1) \quad (14)$$

Putting it all together:

$$S_m = 1 - \sum_{j=2}^{m-1} (\zeta(j) - 1) \quad (15)$$

$$= 1 - \sum_{j=2}^{m-1} \zeta(j) + \sum_{j=2}^{m-1} 1 \quad (16)$$

$$= 1 - \sum_{j=2}^{m-1} \zeta(j) + (m - 2) \quad (17)$$

$$= (m - 1) - \sum_{j=2}^{m-1} \zeta(j) \quad (18)$$

This completes the proof. □

3 Corollary

Evaluating the theorem at $m = 2$ yields the solution to the original limit image:

$$\sum_{x=2}^{\infty} (\zeta(x) - 1) = (2 - 1) - 0 = 1 \quad (19)$$

Evaluating at $m = 4$, for example, yields:

$$\sum_{x=4}^{\infty} (\zeta(x) - 1) = 3 - \zeta(2) - \zeta(3) \quad (20)$$